

Year 8 Science – Booklet 19: Biology

Respiration & Circulation (Breathing) — ANSWERS

2.3 Breathing – In-text questions

- A. It moves up and out (the muscles between the ribs contract, pulling the ribcage up and out).
- B. It relaxes and moves up.

Summary Questions

1. Copy and complete the table (3 marks)

	Inhaling	Exhaling
ribs move	up and out	down and in
diaphragm moves	down	up
chest volume	increases	decreases

2. Two factors that can reduce lung volume (2 marks)

Smoking and diseases such as asthma (also accept: old age).

3. How to measure Usain Bolt's lung volume (3 marks)

Ask him to breathe in as deeply as possible, then breathe out fully and forcefully through a tube into a plastic bottle completely full of water, inverted in a tank of water. As he exhales, the air pushes an equal volume of water out of the bottle into the tank. Measuring the volume of water displaced (e.g. with a measuring cylinder) gives his lung volume.

4. Bell-jar model of inhalation, with diagram and a limitation (6 marks QWC)

In the model, the bell jar represents the chest, the two balloons represent the lungs, and the stretched rubber sheet across the base represents the diaphragm. To model inhaling, the rubber sheet is pulled down – representing the diaphragm contracting and moving down. This increases the volume inside the jar, decreasing the pressure inside it, so air rushes in through the tube and the balloons inflate, representing air being drawn into the lungs.

Diagram: a bell jar with a Y-shaped tube at the top (open to the air) leading to two balloons inside the jar, and a stretched rubber sheet sealing the open base, pulled downward by hand to inflate the balloons.

Limitation: the model doesn't show the ribcage moving up and out during breathing (the jar's rigid walls don't move like a real ribcage), and the rubber sheet is pulled by hand rather than contracting/relaxing like a real diaphragm muscle, so it doesn't fully replicate the actual muscular movements of breathing.

B2.3 Breathing (workbook)

A. Fill in the gaps

When you inhale, the muscles between your **ribs** and your diaphragm **contract**. This **increases** the volume of your chest cavity which **decreases** the pressure, causing air to be drawn in. When you exhale, the muscles **relax**; this **decreases** the volume of your chest cavity. This **increases** the pressure and forces air **out**. You can use a **bell-jar model** to model this process. Smoking and diseases such as **asthma** can reduce lung **volume**.

B. Explain how the ribcage and diaphragm cause you to inhale

The intercostal muscles (between the ribs) contract, pulling the ribcage up and out, while the diaphragm also contracts and moves downward. Together these increase the volume inside the

chest cavity; since the same amount of air now fills a larger space, the pressure inside the chest drops below atmospheric pressure, so air is drawn in from outside (high to low pressure) to fill the lungs.

C. Bell-jar model

- a). Pushing the rubber sheet upward represents the diaphragm relaxing and moving up; this decreases the jar's volume, increasing the pressure inside it, which pushes air out through the tube and deflates the balloons – representing air being forced out of the lungs.
- b). The rigid bell jar doesn't move or change shape like a real ribcage, and the rubber sheet is pulled/pushed by hand rather than contracting and relaxing like a real muscle, so it doesn't fully replicate the actual movements involved in breathing.

D. Using the equipment to measure lung volume

Fill the plastic bottle completely with water and invert it in the water tank, with the plastic tube leading into its mouth. Breathe out as fully as possible through the tube; the exhaled air collects at the top of the bottle and pushes an equal volume of water out into the tank. Measuring the volume of water displaced (e.g. with a measuring cylinder) gives the volume of air breathed out – your lung volume.

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